MIL-P-36577A
17 August 1987
Superseding
MIL-P-36577
17 July 1969

MILITARY SPECIFICATION

PLUGGER, AMALGAM, DENTAL (Markley)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dental pluggers suitable for use in condensing silver amalgam during dental restorative procedures.

1.2 Classification. Pluggers shall be furnished in the style specified for the applicable stock number and item identification, (see 6.1).

Style 1. - No. 1 Markley
Style 2. - No. 3 Markley
Style 3. - No. 4 Markley
Style 4. - No. 5 Markley

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6520

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SPECIFICATIONS

Federal

GG-I-526	- Instruments, Dental and Surgical, General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Military

MIL-P-116	- Methods of Preservation.
MIL-L-10547	- Liners, Case, and sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

Federal

FED-STD-376	- Preferred Metric Units for general use by the Federal Government.

DRAWING

Defense Personnel Support Center (DPSC)

24173 Plugger, Amalgam, Markley.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 135 - Specification for Seamless Brass Tube.
ASTM E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E 92 - Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The plugger shall be in accordance with the applicable requirements of GG-I-526 with the options and additional requirements specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is a specification control drawing. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper function, fit, assembly, or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

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3.2 Material.

3.2.1 Operative ends. The operative ends shall be fabricated from class 4, type 440A martensitic steel.

3.2.2 Handles. Handle A shall be fabricated of low leaded brass, alloy no. 330. Handle B shall be fabricated of cartridge brass, alloy no. 260. Brass shall conform to copper alloy No. 260 or No. 330 as specified in ASTM B135 for chemical composition only. Handle A and B shall be chromium over nickel plated brass. Handle C shall be fabricated of class 3, type 303 stainless steel as specified in GG-I-526. The corrosion tests, as specified in 3.7 shall apply.

3.3 Design and dimensions. The design and dimensions of the pluggers shall conform to DPSC Drawing No. 24173 for the applicable style and number indicated. The amalgam plugger shall be suitable for use in condensing plastic silver amalgam in pin reinforcement techniques.

3.4 Construction. The plugger shall be double-ended and of three piece construction, with the shanks of the operative ends secured in the handle. The interface of the shank and handle shall be smoothly blended with no crevices evident. Handle A shall be octagonal. Handles B and C shall be round. Handles A and B shall be hollow. Handle C shall be solid. The finger grip portions of handles B and C shall be diamond knurled. Handle A shall have superficial diamond serrations on seven faces and shall have one smooth face for identification marking as shown on DPSC drawing 24173.

3.5 Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C52 to C56. Handles and shanks shall be properly tempered and shall have a hardness suitable for the purpose intended. Hardness shall be determined as specified in 4.4.1.

3.6 Finish. The operative end including shank shall have a smooth mirror or smooth bright finish. The operative end shall be free of defects such as grind marks, cracks gouges, burrs, feathers or other imperfections. Handles shall have a mirror or bright finish, except for the diamond knurl. Finger grip or serrated portion of handle shall be free from barbs and shall provide adequate finger grip during use.

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3.7 Corrosion resistance. The entire instrument shall pass the boil test specified in GG-I-526. The copper sulfate test requirements of GG-I-526 shall apply to the entire plugger for handle C (corrosion-resisting steel handle) but only to the operative ends for pluggers with handles A and B (chromium over nickel plated brass).

3.8 Identification marking. The smooth side of the octagonal handle and the middle section of the round handles shall be permanently and legibly marked in accordance with GG-I-526. In addition to the contractor marking, each plugger shall be marked with the name or trademark of the manufacturer, if different from the contractor. The word "stainless," when applicable, shall also be marked on the pluggers of foreign manufacture. The instrument number "1M","3M", "4M", and "5M" as applicable, shall be imprinted on round handles in middle section within 1/2 inch of diamond knurl and within 1/2 inch of the lengthwise edge of the smooth side of octagonal handles.

3.9 Workmanship. The plugger shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

4.2 Quality conformance inspection. The examinations required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in Tables I and III (see 4.2.1 and 4.3). The tests required are specified in Table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in Table I. The unit of product for examination purposes shall be one plugger.

TABLE I. Sampling for examination.

Type defect	Inspection Level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one instrument.

TABLE II. Sampling criteria for tests.

Characteristic	Requirement	Test Procedure	Inspection level	AQL
Hardness	3.5	4.4.1	S-2	1.5
Copper Sulfate Test	3.7	GG-I-526	S-2	1.0
Boil Test	3.7	GG-I-526	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of raw material, used in the fabrication of the operative ends and the handle of the plugger, shall be selected for the chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Examination. The pluggers shall be examined to determine compliance with the requirements contained in this specification. Any plugger in the sample containing one or more defects shall be rejected, and if the number of defective pluggers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The pluggers shall be examined for defects including, but not limited to, those listed in Table III.

TABLE III. Classification of defects.

Category	Defect
Major A:	
101	Plugger not of applicable style and number.
102	Operative end not free of grind marks, cracks, burrs, or feathers.
103	Plugger not free from cracks.
104	Shanks not properly secured in the handle.
105	Finish on operative ends and shank not mirror or not bright polished.

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TABLE III. Classification of defects.

Category	Defect
Major B:	
151	Dimensions of operative ends not as specified.
Minor A:	
201	Handle serrations of incorrect type, or missing altogether.
202	Finish on handle not as specified.
203	Plugger not free from foreign matter such as dirt, oil, or polishing compound.
204	Identification marking not present or not permanent, legible, or complete.
205	Identification marking, or any portion thereof, at incorrect location.
Minor B:	
251	Any dimension (other than those in 151) not as specified.

4.4 Test procedures. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.2).

4.4.1 Hardness. Hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1. Level A.

5.1.1.1 Cleaning. Plugger shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Plugger shall be dried in accordance with MIL-P-116.

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5.1.1.3 Preservative applications. Preservatives shall not be used.

5.1.1.4 Unit package. Each plugger shall be packaged in accordance with MIL-P-116, method IC-1. As an alternate, each plugger shall be packaged in a rigid plastic tube having a metal or plastic screw cap or a plastic plug type closure. The plugger shall be immobiiized within the container and the operative ends shall be protected with cork or other suitable material.

5.1.1.5 Intermediate package. Twelve unit packages containing pluggers of one style only shall be packaged in a snug fitting box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for pluggers shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1.)

5.2.1 Level A. Twenty-four intermediate packages containing pluggers of one style only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B 636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twenty-four intermediate packages containing pluggers of one style only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for pluggers shall conform to the MIL-STD-794 requirements for this level.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallets loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to type IV or type V.

5.2.5.1 Levels A and B. Pluggers packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Pluggers packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2 and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, date of this specification.
(b) Style required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN)/coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Style
6520-00-935-7167	PLUGGER, AMALGAM, DENTAL, Markley, No. 1	1
6520-00-935-7168	PLUGGER, AMALGAM, DENTAL, Markley, No. 3	2
6520-00-935-7169	PLUGGER, AMALGAM, DENTAL, Markley, No. 4	3
6520-00-935-7170	PLUGGER, AMALGAM, DENTAL, Markley, No. 5	4

6.3 Subject term (key word) listing:
Amalgam
Dental
Markley
Plugger

MILITARY INTERESTS:

Custodians:
Army-MD
Navy-MS
Air Force - 03

Review Activities:

Army-MD
Navy-MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

05460

Project No.6520-1858

Location: NCR DisK #2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-36577A	PLUGGER, AMALGAM, DENTAL (Markley)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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